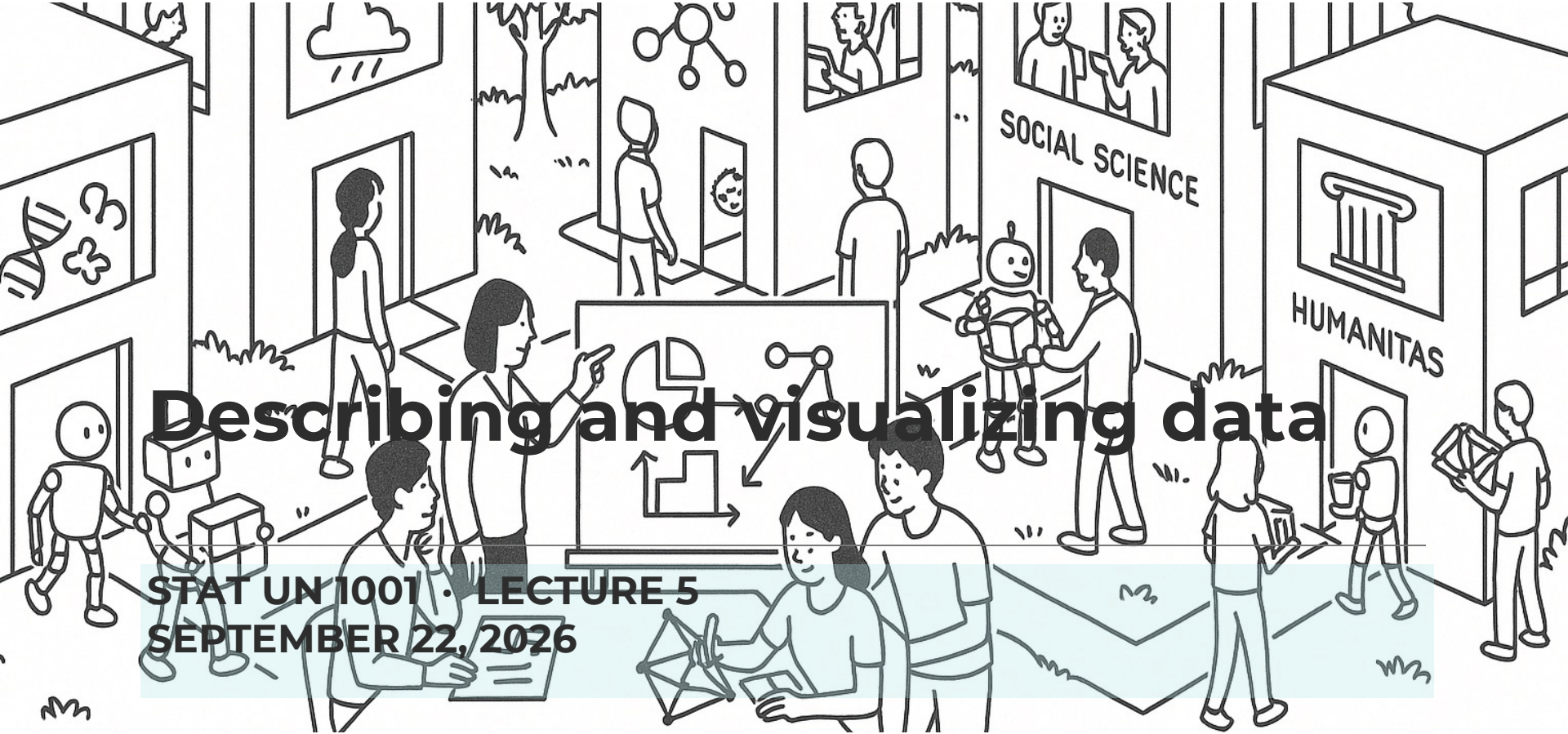


Describing and visualizing data

STAT UN 1001 · LECTURE 5
SEPTEMBER 22, 2026

Week 3 plan

NUMERICAL DATA

Mean notation, 0/1 variables, bins, quartiles, variance, standard deviation, shape, and transformations

CATEGORICAL DATA

Contingency tables, conditional proportions, association, and comparison

Interpreting Graphs (Figures)

Bar plot, histogram, boxplot, scatterplots, other plots in the book

Thursday: project discussion and work session

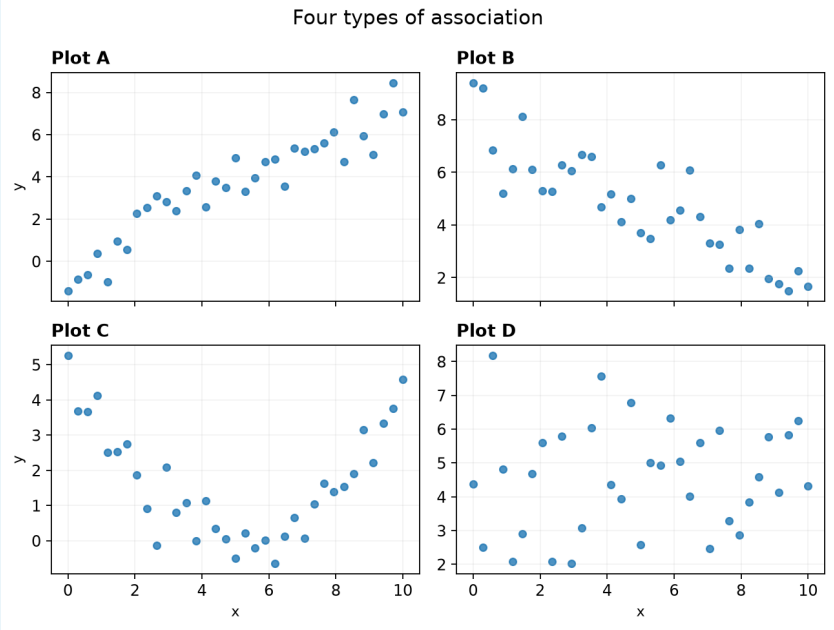
What the flipped activity showed

Binned counts	**61.9%**
Quartiles	**73.0%**
Column proportion	**76.2%**
Row proportion	**77.8%**
0/1 sum	**81.0%**

38.7%

reported some uncertainty about frequency tables, contingency tables, or conditional proportions

Quiz questions 1–2 | Association



Which classification is correct?

A. A positive linear, B negative linear, C nonlinear, D no clear association

B. A negative linear, B positive linear, C no association, D nonlinear

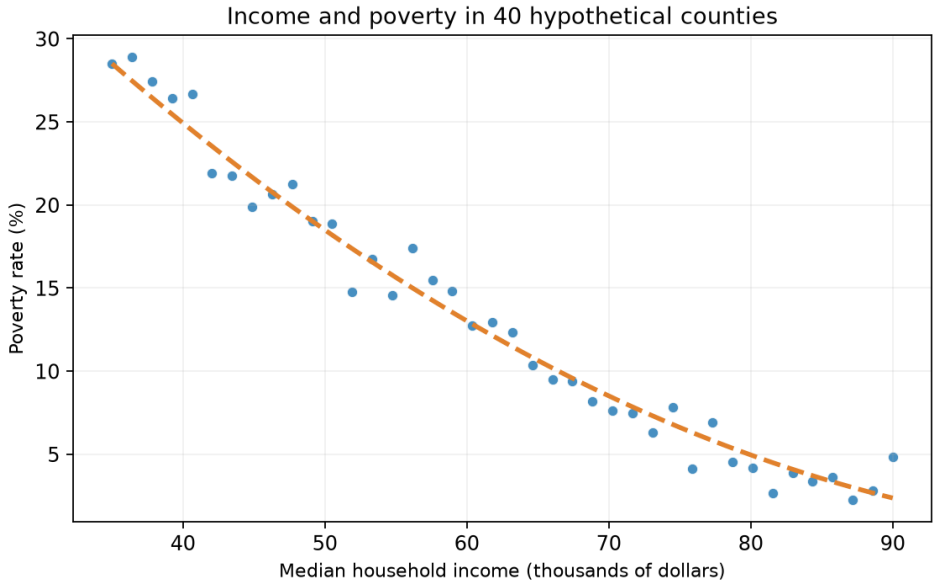
C. A and B nonlinear, C and D linear

D. All four show positive linear association

What makes plot C nonlinear?

Association and nonlinearity

Q1: 96.8% correct



A straight line assumes a nearly constant slope.

The pattern decreases steeply at first and then flattens. Its slope changes across the x-axis.

Strong association can be nonlinear.

Quiz questions 3–4 | The mean

In $\bar{x} = (x_1 + x_2 + \ldots + x_n) / n$, what does n represent?

A. The number of observations in the sample

B. The numerical value of the final observation

C. The population mean

D. The number of variables in every data set

X is the variable name;

x_1 is the first value for case #1

Mean is a summary of all observed values of this variable

Compute the mean of 1, 2, 1, 2, 4.

Mean notation in one example

Q3: 84.1% correct

$x_1 + x_2 + \ldots + x_n$

the observed values

n

the number of observations

$$1 + 2 + 1 + 2 + 4 = 10$$

$$10 / 5 = 2$$

Quiz questions 5–6 | A 0/1 variable

Eight service requests

1, 0, 1, 1, 0, 0, 1, 1

1 means resolved within one day. 0 means otherwise.

Cases?
Variable?
Values?

Q5. What does the sum represent?

A. The number resolved within one day: 5

B. The number not resolved: 3

C. The total number of requests: 8

D. The average resolution time: 5

Q6. What does the mean represent?

A. 5/8 = 0.625, the proportion coded 1

B. 3/8 = 0.375, the proportion coded 1

C. 5, the average number of days

D. 8/5 = 1.6, requests per day

The 0/1 indicator rule

Q5: 81.0% correct

1 + 0 + 1 + 1 + 0 + 0 + 1 + 1

SUM

5

number of cases coded 1

MEAN

5/8 = 0.625

fraction of cases coded 1

The key is to understand what the denominator should be: a mean of what values among which cases?

Quiz question 7 | Create binned counts

61.9% correct

Values

5.2, 7.4, 7.5, 8.1, 9.9, 10.0, 10.8, 12.5

This is confusing. Will avoid in the future.

Bins

5.0–7.5 **7.5–10.0** **10.0–12.5**

Place a boundary value in the lower bin. What are the three counts?

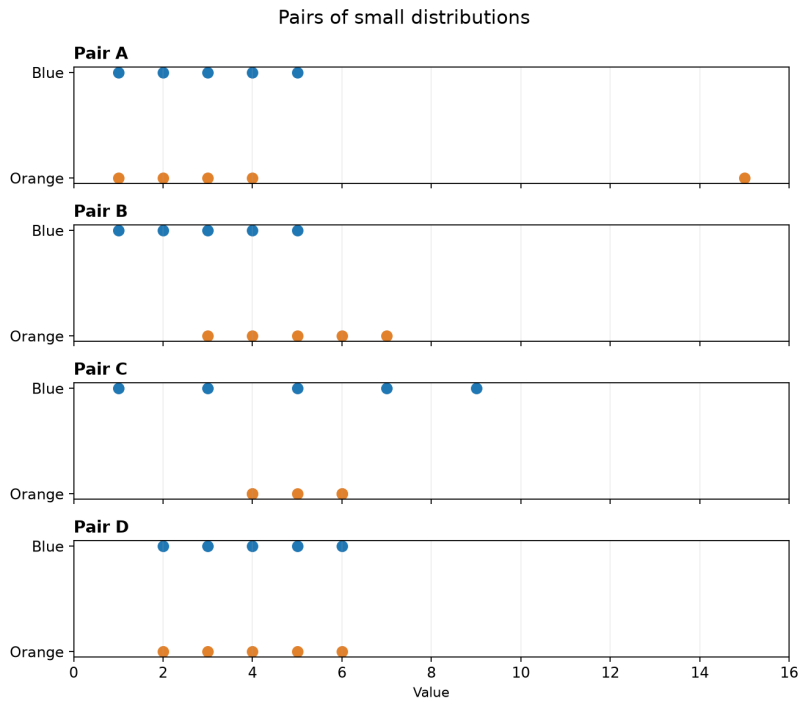
A. 3, 3, 2
B. 2, 3, 3
C. 2, 2, 4
D. 3, 2, 3

Boundary values determine the counts

5.0–7.5	7.5–10.0	10.0–12.5
5.2 7.4 7.5	8.1 9.9 10.0	10.8 12.5
3	**3**	**2**

A histogram begins with a counting rule that assigns every observation to one bin.

Quiz questions 8–9 | Center and spread



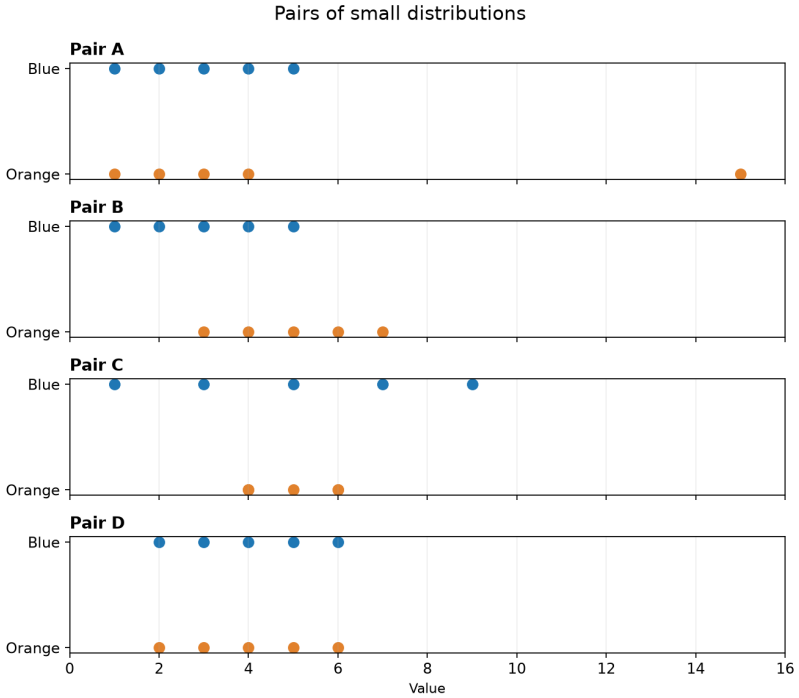
Q8

Which pair has equal medians and a larger blue IQR?

Q9

For Pair A, which color has the larger mean and standard deviation?

Center and spread answer



Q9: 81.0% correct

Q8: Pair C

Both medians are 5. Blue has the wider middle 50%.

Q9: Orange

The value 15 moves the mean right and increases the distances from the mean.

Quiz question 13 | Median and quartiles

73.0% correct

Ordered values

1 2 3 4 5 6 7 8 9

Exclude the median when finding the lower and upper quartiles. What are the median, Q1, and Q3?

A. Median = 5, Q1 = 2.5, Q3 = 7.5

B. Median = 5, Q1 = 3, Q3 = 7

C. Median = 4.5, Q1 = 2, Q3 = 8

D. Median = 5, Q1 = 1, Q3 = 9

It is like cutting a cake.

Quartiles are medians of the two halves

1 2 3 4 — lower half

5 — median

6 7 8 9 — upper half

Q1 = (2 + 3) / 2 = 2.5

Q3 = (7 + 8) / 2 = 7.5

IQR = Q3 − Q1 = 5

Variance and standard deviation

Dataset A

2 2 2 2

mean = 2 SD = 0

Dataset B

0 2 2 4

mean = 2 SD > 0

Variance

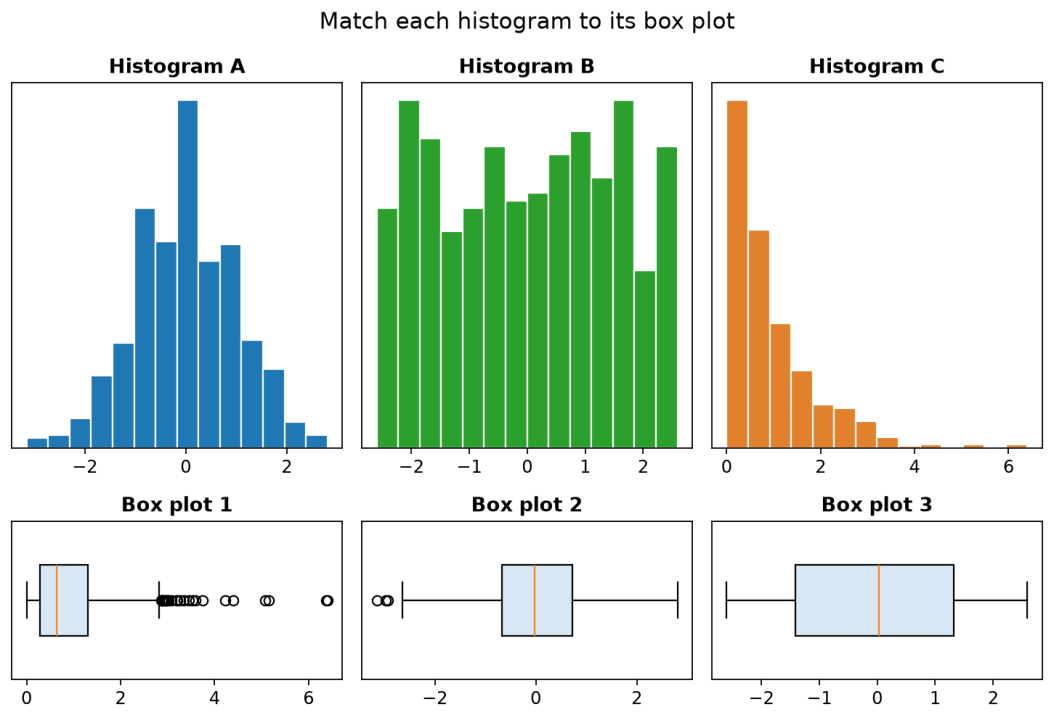
Average squared distance from the mean

Standard deviation

Square root of variance, returned to the original units

Sample SD uses n − 1 because estimating the mean from the same sample makes spread look slightly too small.

Quiz question 10 | Match the displays



Match each histogram to its box plot.

A. A → 2, B → 3, C → 1

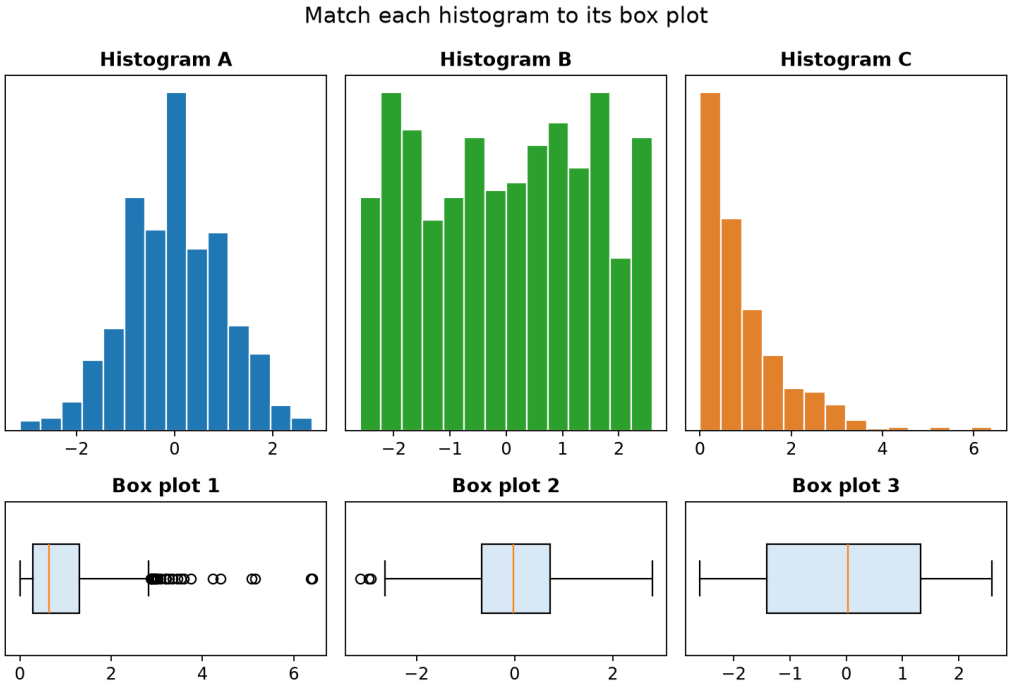
B. A → 1, B → 2, C → 3

C. A → 3, B → 1, C → 2

D. A → 2, B → 1, C → 3

Which visual clues did you use?

Shape, box plots, and modes



A → 2

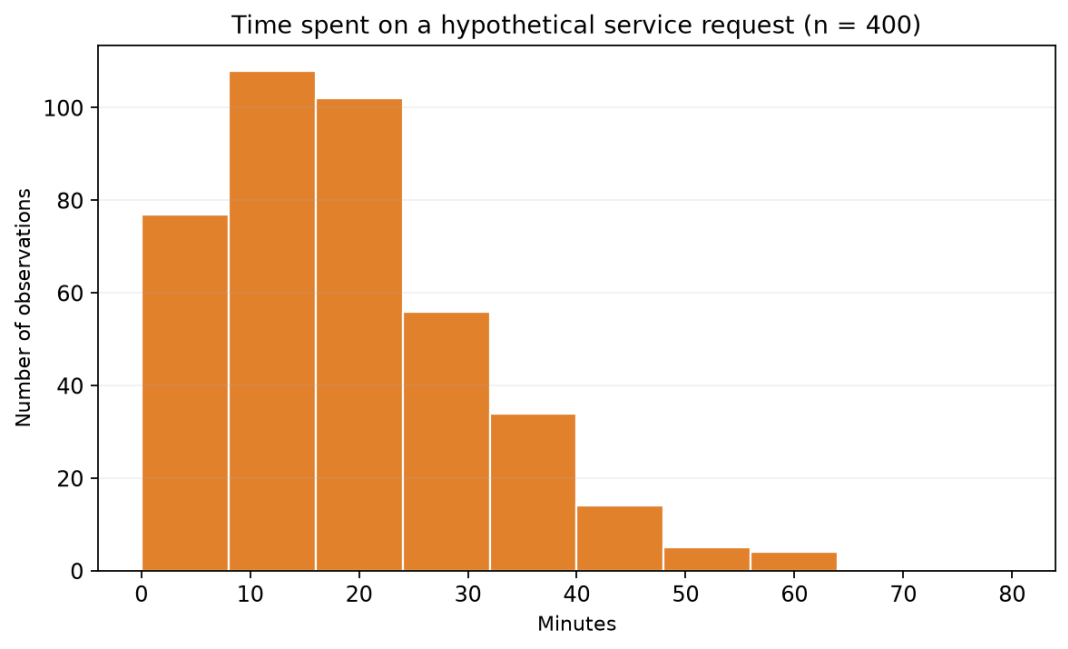
B → 3

C → 1

Quiz question 12

A mode is a prominent peak or concentration, not every one-bin fluctuation.

Quiz question 11 | Read a skewed histogram



A. Median roughly 16–20 minutes, mean probably higher

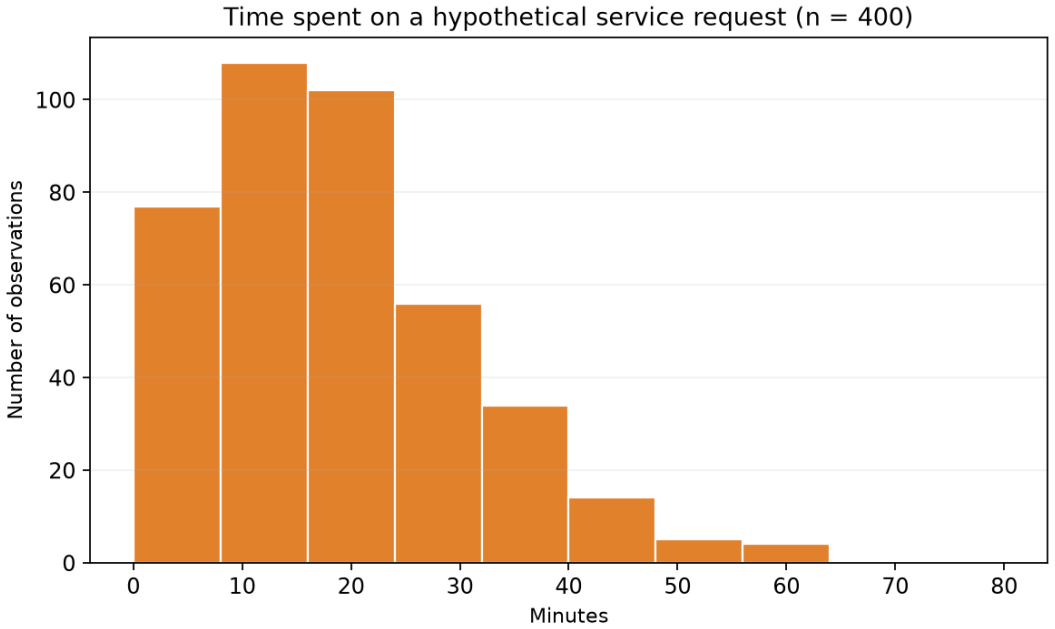
B. Median roughly 55–60 minutes, mean lower

C. Median must equal mean because n = 400

D. Median cannot be estimated from a histogram

Where do about 200 observations fall below?

Skew connects shape to center



Median

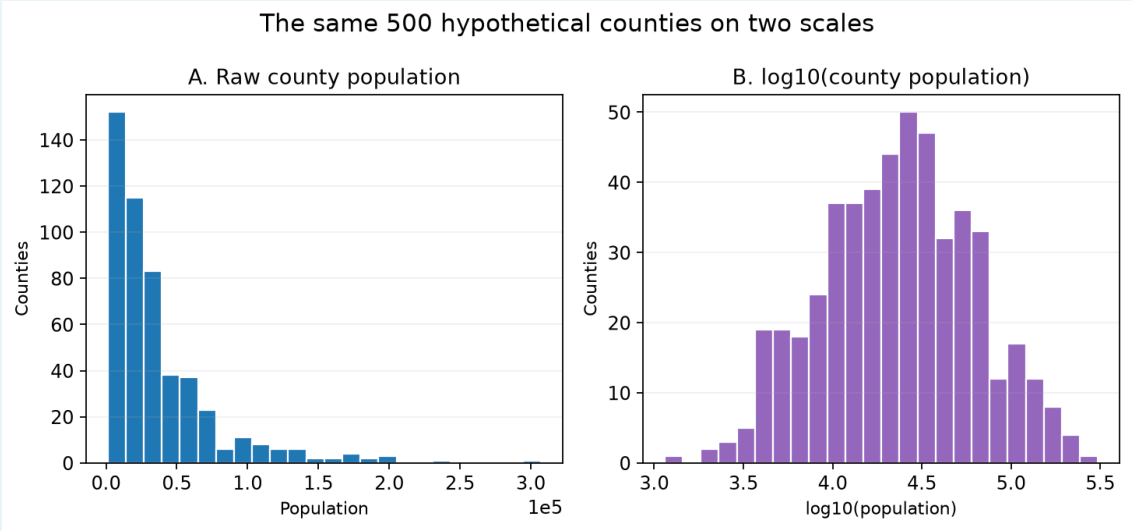
Around 16–20 minutes

Mean

Probably higher than the median

The tail points toward the larger values that pull the mean.

Quiz question 14 | Why use a log scale?



What is the main benefit of panel B?

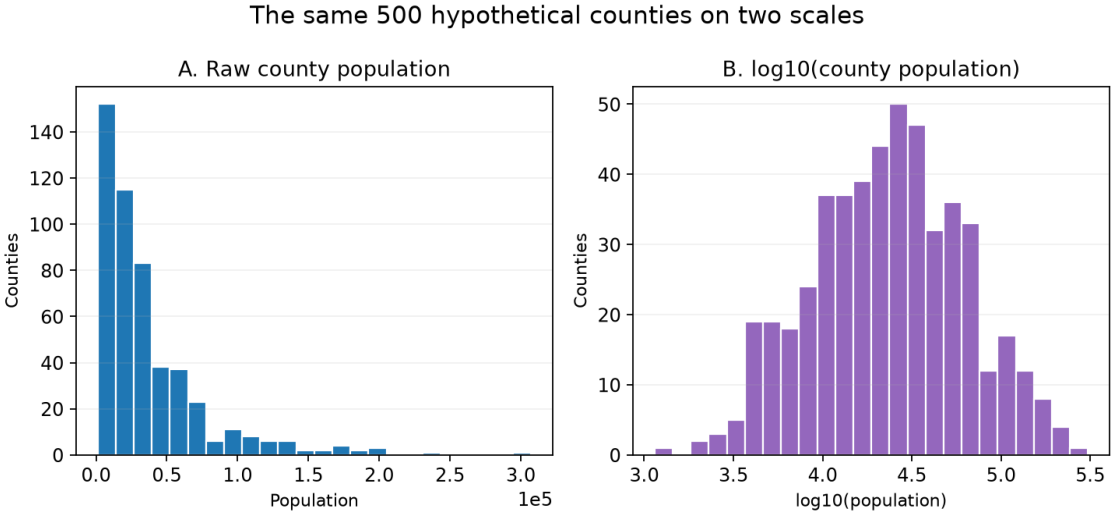
A. Compresses large values and separates smaller ones

B. Removes counties with large populations

C. Makes all counties have the same population

D. Proves the original distribution was symmetric

The log scale reveals crowded structure



Same counties

Different horizontal scale

Large values compress. Smaller values separate. Patterns hidden near zero become visible.

Transformations affect how distances look.

Quiz question 15 | Create a contingency table

85.7% correct, 6 requested the answer

Eight students used a video or text explanation, then rated it helpful.

Video–Yes, Video–Yes, Video–No, Video–Yes
Text–Yes, Text–No, Text–No, Text–No

Entries are ordered as Video–Yes, Video–No; Text–Yes, Text–No. Which table is correct?

A. 3, 1; 1, 3

B. 4, 0; 0, 4

C. 3, 3; 1, 1

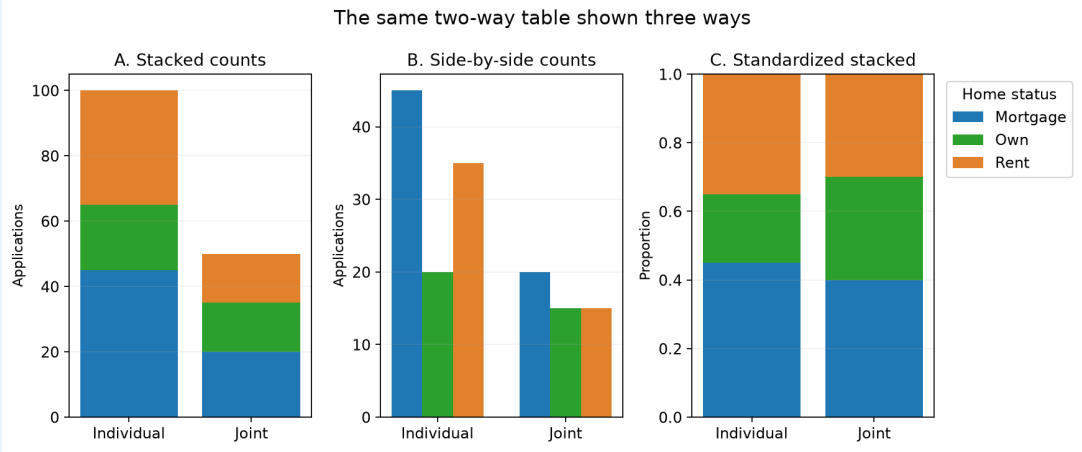
D. 1, 3; 3, 1

Each case enters one cell

Explanation	Helpful: Yes	Helpful: No
Video	3	1
Text	1	3

Row totals answer questions among explanation formats.
Column totals answer questions among helpfulness groups.

Quiz questions 16–17 | Which denominator?



Q16

What does 45/100 represent?

Q17

What does 45/(45 + 20) represent?

Say the denominator in words before calculating.

Conditional proportions begin with “among”

Q16–17: 76–78% correct

45 / 100

Among individual applications, what fraction had a mortgage?

The denominator is a row total.

45 / (45 + 20)

Among mortgage applications, what fraction were individual?

The denominator is a column total.

Condition first. Calculate second.

Quiz question 18 | Association from proportions

Video users	Text users
3 of 4 helpful = 75%	1 of 4 helpful = 25%

Which conclusion is supported descriptively?

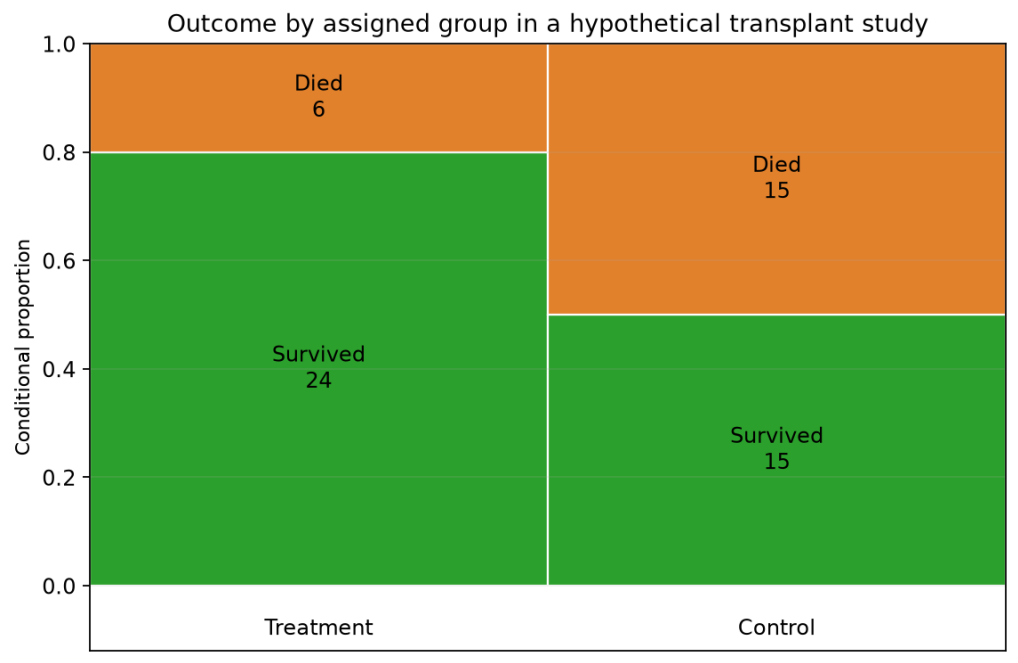
A. Helpfulness was associated with format in this small group

B. The variables were independent because both groups had four students

C. The helpful percentages were the same

D. The table proves video causes greater learning for every student

Quiz question 19 | Randomized treatment comparison



Treatment: 24 survived, 6 died
Control: 15 survived, 15 died

Which comparison is correct?

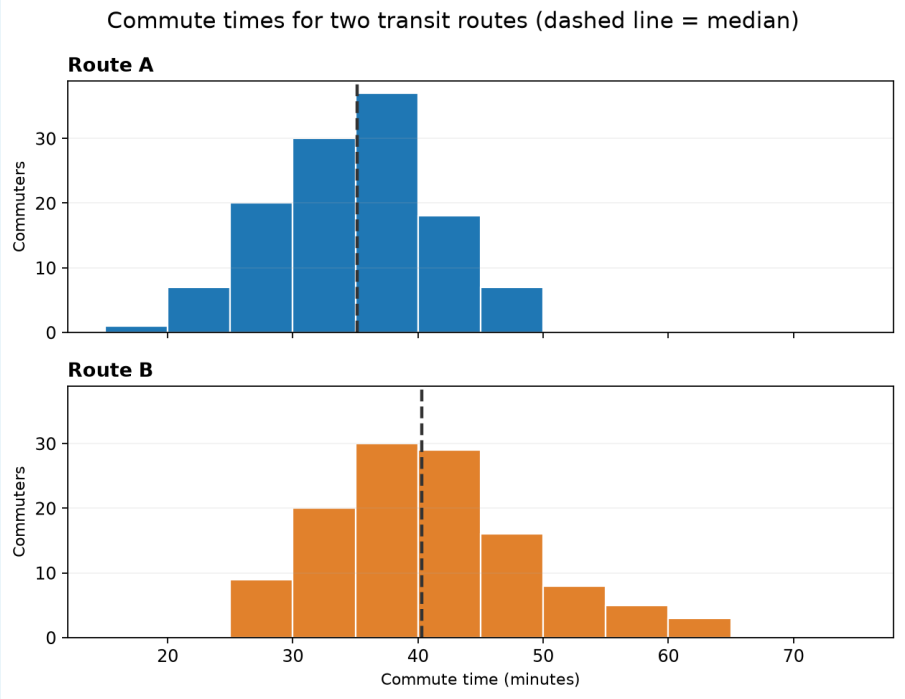
A. 80% versus 50%. Random assignment supports a causal comparison

B. 50% versus 80%

C. The proportions are identical because both groups have 30 patients

D. No proportion can be computed from a mosaic plot

Quiz question 20 | Compare distributions



Which comparison is best supported?

A. Route A has a lower typical time and less variability

B. Route A has a higher typical time and more variability

C. The routes have identical centers and spreads

D. Only the maximum matters

Center · spread · shape · unusual observations

One-minute exit check

1 **Explain the difference between a count and a conditional proportion in one sentence.**

2 **Name one Chapter 2 idea that you can now explain more clearly.**

Thursday is reserved for project discussion and work.

Quiz problems and plots adapted from OpenIntro Statistics, Chapter 2, CC BY-SA 3.0.

NEWS STORY · SEPTEMBER 2026

When the comparison group changes, the question changes

THE BMJ · INVESTIGATION

A vaccine-safety algorithm lost sensitivity when reports for the two new mRNA vaccines came to dominate the database.

More than 90% of roughly 1.1 million first-year VAERS reports concerned the mRNA covid vaccines, according to the investigation.

Source: David Willman, The BMJ, 16 Sep 2026

Start here

What is a VAERS case?

A submitted report of an event after vaccination—not a randomly sampled vaccinated person.

What can a signal do?

Flag a pattern for further study. It does not, by itself, establish causation.

HYPOTHETICAL TEACHING EXAMPLE

Scenario 1: compare Vaccine A with an established baseline

Reported outcome

Vaccine group	Event X	Other events	Total	Event X proportion
Vaccine A	50	950	1,000	5%
Established vaccines	90	8,910	9,000	1%

PRR

5% ÷ 1%

= 5.0

Question answered

Does Vaccine A stand out relative to the reporting pattern for established vaccines?

The cases here are reports; 5% is not the risk among all vaccinated people.

SAME 5% FOR VACCINE A

Scenario 2: another new vaccine dominates “everyone else”

Vaccine group	Event X	Other events	Total	Event X proportion
Vaccine A	250	4,750	5,000	5%
New Vaccine A2	225	4,275	4,500	5%
Older vaccines	5	495	500	1%

“Everyone else”

$$\frac{(225 + 5)}{(4{,}500 + 500)}$$

= 4.6%

PRR = 5% ÷ 4.6% = 1.09

What changed?

Not Vaccine A's 5%. The comparison group changed—from an older baseline to a pool dominated by Vaccine A2.

A statistic is an answer—but to which question?

Interpret the association

PRR ≈ 1 says:

Vaccine A and this particular comparison group look similar in the reporting database.

It does not say:

Both match an established baseline—or that no causal risk exists.

Redesign the comparison

1. **What population-level question do we want to answer?**
2. Which reports belong in a meaningful baseline group?
3. Should we stratify by age, sex, or time period?
4. What other surveillance data could corroborate a signal?

Discussion: Write one sentence stating the question answered by each table.